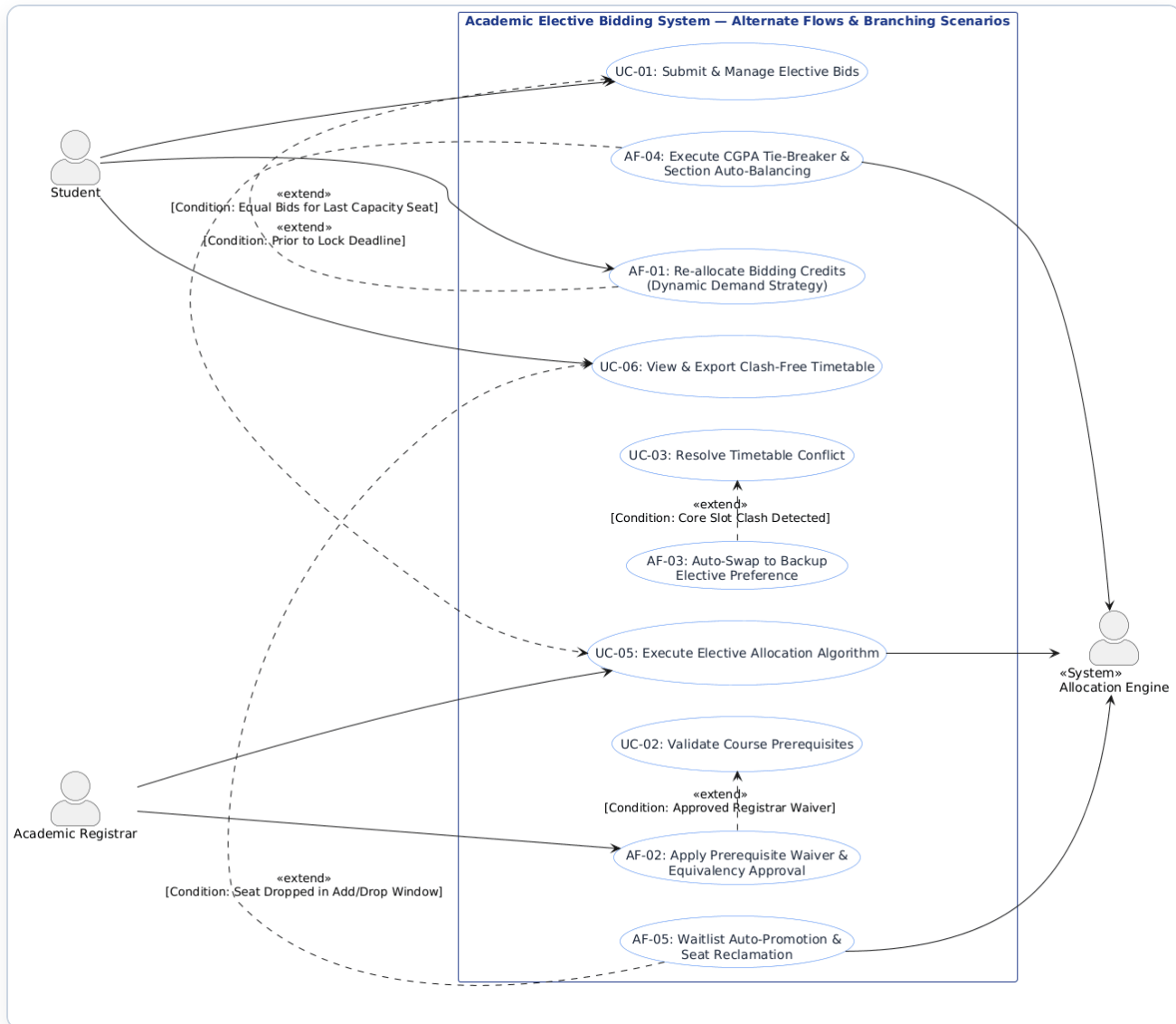


Deliverable 4: Alternate Flow Use-Case Diagram & Specifications

Course: Software Engineering Lab 1 (PES University, Dept. of CSE)

Student: Shashank D | **SRN:** PES1UG24CS433

Problem Statement #05: Academic Elective Bidding & Allocation System



Alternate Flow Specifications

- **AF-01: Re-allocate Bidding Credits (Dynamic Demand Strategy - «extend» UC-01)**
Condition: Prior to Bidding Deadline Closure.

Flow: Student reviews dynamic seat popularity metrics; adjusts points across draft electives to maximize allocation likelihood; saves updated 100-credit distribution.

- **AF-02: Apply Prerequisite Waiver & Equivalency Approval («extend» UC-02)**

Condition: Registrar Waiver Granted for Course Prerequisites.

Flow: Student lacks standard prerequisite but presents signed registrar waiver token; system bypasses standard check and approves bid submission.

- **AF-03: Auto-Swap to Backup Elective Preference («extend» UC-03)**

Condition: Primary Elective Timetable Clash Detected.

Flow: When top-ranked preference conflicts with core slot, engine automatically shifts evaluation to pre-ranked secondary backup course without terminating submission.

- **AF-04: Execute CGPA Tie-Breaker & Section Auto-Balancing («extend» UC-05)**

Condition: Equal Bid Points for Final Capacity Seat.

Flow: Solver breaks point ties using Cumulative GPA and prerequisite grade index, assigning tied student to parallel open section slot.

- **AF-05: Waitlist Auto-Promotion & Seat Reclamation («extend» UC-06)**

Condition: Seat Vacated during Add/Drop Window.

Flow: Engine reclaims vacated seat and automatically promotes highest waitlisted student, updating timetable and dispatching notification.